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DEPARTMENT OF PHYSICS AND TECHNOLOGY

PHYS299

Bachelor Project in Physics

Modelling the Energy Levels of Atoms using the Hartree-Fock Method on a Radial Finite Difference Grid.

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The Hartree–Fock method provides an approximate approach for determining the energy levels and total energy of atomic systems. It employs a combination of theoretical formalism and numerical techniques to address cases where analytical solutions are not available. This report presents a study on firstly Lithium following onto Beryllium, Boron and Carbon’s energy levels using the Fock matrix method. It is intended that theory and the accompanying code may serve as a resource for future students, helping to bridge the gap between the theoretical framework and practical implementation of approximate quantum methods.

1 Introduction

The Schrödinger equation for hydrogenic systems is typically introduced during the early stages of undergraduate study, where it can be solved analytically to obtain exact energy levels. However, for multi-electron atoms the problem becomes significantly more complex. For example with Helium, although approximate treatments can reduce the system to an effective two-body problem, an exact analytical solution is no longer attainable, necessitating the use of numerical methods. As the number of electrons increases, the complexity of the governing equations grows rapidly due to electron–electron interactions. In systems containing more than two electrons, additional exchange terms arise for electrons with parallel spin, reflecting the antisymmetry requirement of the total wavefunction imposed by the Pauli exclusion principle. This leads to observable effects on electron binding and energy levels. The aim of this project was to introduce a Hartree Fock method by following the textbook by Szabo and Ostlund [1], and to provide a bridge between this and the steps to reproducing a numerical method. First the theory and how we approach these problems must be covered. Then the report will discuss how to code the problem and the differences between the closed and open shells resulting in the restricted and un-restricted model. And then the different procedure for Boron as we diverge from a spherical approximation is covered. Finally the results and accuracy of the code will be discussed. The atoms this report covers is that of Lithium, Beryllium, Boron and Carbon by extension it should be possible to model all atoms if the orbital shapes are modeled as spheres which is what is done in Boron and Carbon, these examples will be discussed.

2 Theory

2.1 Many body problem and Hamiltonian

This project looks to solve the energy levels of atoms larger than helium, we will be working in atomic units, the Hamiltonian that is the focus of the study is such:

$$\hat{H} = - \sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{i=1}^N \frac{Z}{r_i} + \sum_{i=1}^N \sum_{j>i}^N \frac{1}{r_{ij}} \quad (1)$$

This operator consists of three main components: the kinetic energy of each electron, the Coulomb attraction between the positively charged nucleus and each electron, and finally, the electron-electron interaction. The last term is what introduces the primary challenge of the many-body problem, as the $1/r_{ij}$ term intricately couples the motions of all electrons together.

To find the exact energy levels of the atom, we must solve the time independent Schrödinger equation.

$$\hat{H}\Psi = E\Psi \quad (2)$$

Because the electron-electron repulsion term makes this equation analytically unsolvable for systems with more than one electron, we must rely on advanced approximation methods. The model we will use is the central finite difference approximation, and we reduce the wavefunction into separate spherical and radial components. It is possible to solve just the radial equation due to spherical symmetry. This can be done through the central field approximation which is an assumption that every electron moves independently in an average potential. Meaning each electron sees a central potential that is only a function of r (radial distance). This is possible as the radial and spherical components can be separated to satisfy their own equations [2]. This is not to say we only deal with spherical potentials, when we move from the s -orbitals to the first p -orbits as in Boron the electron shells are a dipole shape, we deal with this with a slightly different term in the multipole expansion seen in section 2.5. Ignoring the electron-electron term for now, this allows us to reduce our Eq.(1) to [3]:

$$\left[-\frac{1}{2} \frac{d^2}{dr^2} + V(r) + \frac{l(l+1)}{2r^2} \right] u(r) = \epsilon u(r) \quad (3)$$

Now we are able to use a radial grid to solve this equation. In this expression, the kinetic energy operator is simplified to a second-order derivative with respect to r , which can be mapped directly onto a finite difference grid as seen with Eq.(4):

$$\frac{d^2 u}{dr^2} = \frac{u_{i-1} - 2u_i + u_{i+1}}{r^2} \quad (4)$$

This has a simple matrix form of -2 along the diagonal and +1 on either side - a triangular matrix [4].

When solving these problems computationally we transition from continuous wavefunctions to a discrete set of points, following the Roothaan-Hall approach as outlined by Baskerville [5], allowing for the use of standard linear algebra eigensolvers. This is represented by our radial grid a vector, which we can write as a matrix being necessary later. The second term in Eq.(3), $V(r)$ represents the attractive nuclear potential, $-Z/r$, and the final term, $\frac{l(l+1)}{2r^2}$, represents the centrifugal term. This is the angular momentum representation, this being dependant on l only needs to be considered for larger atoms such as Boron and Carbon for our study. It introduces a repulsive barrier near the nucleus. This physically prevents electrons with higher angular momentum from penetrating the core as effectively as s-electrons. By defining the Hamiltonian in terms of the reduced radial wavefunction $u(r)$, the numerical task becomes the diagonalization of a triangular matrix where the diagonal elements contain the potential and centrifugal terms at each grid point r_{ij} . This approach streamlines the computation and shows the $1/r^2$ dependence of the centrifugal term and the necessity of a high point density near the origin. The electron-electron interaction term is a bit more complicated than the above so it is essential that more theory is covered before coming to the terms which represent this. Once this is done we can combine our matrices to solve for the energy levels of the system.

2.2 The wavefunction and the Slater determinants

The last term from Eq.(1) needs to be represented. The wavefunction we are interested in must obey the Pauli exclusion principle which can be seen in the results of the energies of the different electrons. Electrons with the same spin can not occupy the same area of space so on average the electrons will be farther from each other which can reduce or increase the energy of the electron depending on the case, this has to be considered alongside the 'standard' electron-electron coulomb repulsion. As we would expect there is a repulsion term between the two electrons we look at for a given time. There is also going to be this additional second term to consider due to Pauli's exclusion principle. The anti-symmetry requirement of the multi-electron wavefunction is satisfied by the use of a Slater Determinant. [6], an example is shown below in Figure [1]. This would be the case for Lithium, therefore we build a 3x3 matrix as there are 3 electrons in the system, we must consider both the spin and position.

The Slater determinant ensures that the many-

electron wavefunction is completely antisymmetric with respect to the exchange of any two particles. Mathematically, swapping the coordinates of two electrons is equivalent to swapping two rows or columns in the determinant, which multiplies the entire wavefunction by -1.

In this implementation, rather than calculating the full determinant explicitly, we utilise the resulting physics, the separation of electron interactions into two distinct types. The Coulomb term represents the classical electrostatic repulsion between electron charge densities, while the Exchange term is a purely quantum mechanical effect arising from the antisymmetry requirement. These terms form the foundation of the Fock operator, allowing the many-body problem to be treated as a series of single-particle equations.

2.3 Deriving the Hartree Fock equations using the variational principle and Lagrange multipliers

The derivation for the Hartree Fock equations will be shown now and follow close to [1] method to ensure a good understanding of the problem. First we will use the variational principle to minimise the energy expression for a Slater determinant, later we shall look at the difference when introducing assumptions around spin and how this will allow us to solve the equations we are building. The variational principle allows us to start with a trial function and minimise this point, it will always be greater than that of the actual value we are interested in but will give a close value. To approximate the electronic wave function Φ , we expand it as a linear combination of a set of N known basis functions $\{\Psi_j\}$:

$$\Phi = \sum_{j=1}^N c_j \Psi_j \quad (5)$$

Our goal is to minimise the energy expectation value $E = \langle \Phi | H | \Phi \rangle$ subject to the normalization constraint $\langle \Phi | \Phi \rangle = 1$. Using the method of Lagrange multipliers, we define the functional $\mathcal{L}$:

$$\mathcal{L} = \langle \Phi | H | \Phi \rangle - E(\langle \Phi | \Phi \rangle - 1) \quad (6)$$

Substituting our linear expansion into the functional gives:

$$\mathcal{L} = \sum_{i,j} c_i^* c_j \langle \Psi_i | H | \Psi_j \rangle - E \left(\sum_{i,j} c_i^* c_j \langle \Psi_i | \Psi_j \rangle - 1 \right) \quad (7)$$

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3) = \begin{vmatrix} \psi_{1s}\alpha(\mathbf{r}_1) & \psi_{1s}\beta(\mathbf{r}_1) & \psi_{2s}\alpha(\mathbf{r}_1) \\ \psi_{1s}\alpha(\mathbf{r}_2) & \psi_{1s}\beta(\mathbf{r}_2) & \psi_{2s}\alpha(\mathbf{r}_2) \\ \psi_{1s}\alpha(\mathbf{r}_3) & \psi_{1s}\beta(\mathbf{r}_3) & \psi_{2s}\alpha(\mathbf{r}_3) \end{vmatrix}$$

Figure 1: The Slater determinant for lithium, with α representing up spin and β representing down spin.

By defining the Hamiltonian matrix elements $H_{ij} = \langle \Psi_i | H | \Psi_j \rangle$ and the overlap matrix elements $S_{ij} = \langle \Psi_i | \Psi_j \rangle$, we require the first variation of $\mathcal{L}$ with respect to the coefficients c_i^* to vanish as $\delta_{c_i^*} \mathcal{L} = 0$. This leaves us with the set of equations:

$$\sum_{j=1}^N (H_{ij} - E S_{ij}) c_j = 0 \quad (8)$$

In matrix form, this results in the generalized eigenvalue problem:

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c} \quad (9)$$

Now moving onto spin orbitals we minimise the energy again but this time with a different constraint, the spin orbitals,

To determine the optimal spin orbitals $\{\chi_a\}$, we minimise the electronic energy E_0 subject to the orthonormality constraint $\int dx_1 \chi_a^*(1) \chi_b(1) = \delta_{ab}$. We introduce a Lagrangian $\mathcal{L}$ with a matrix of Lagrange multipliers ϵ_{ba} :

$$\mathcal{L}[\{\chi_a\}] = E_0[\{\chi_a\}] - \sum_{a=1}^N \sum_{b=1}^N \epsilon_{ba} \left(\int dx_1 \chi_a^*(1) \chi_b(1) - \delta_{ab} \right) \quad (10)$$

where E_0 is the expectation value of the single Slater determinant. This links directly to our Hamiltonian and what we will be solving for numerically, the first term is the kinetic and nucleus attraction, the second term is the coulomb term and the exchange term:

$$E_0 = \sum_{a=1}^N \langle a | h | a \rangle + \frac{1}{2} \sum_{a,b}^N ([aa|bb] - [ab|ba]) \quad (11)$$

Requiring the first variation of $\mathcal{L}$ to vanish ($\delta\mathcal{L} = 0$) leads to the Hartree-Fock integro-differential equations, notice we are left with coupled equations:

$$f(1)\chi_a(1) = \sum_{b=1}^N \epsilon_{ba} \chi_b(1) \quad (12)$$

Here, $f(1)$ is the Fock operator, defined as the sum of the core-Hamiltonian $H(1)$, which contains the kinetic energy and the nuclear potential of the atom, and the effective potential generated by all occupied orbitals:

$$f(1) = H(1) + \sum_{b=1}^N (J_b(1) - K_b(1)) \quad (13)$$

The (1) denotes the coordinates of a single electron. The Coulomb operator J_b and Exchange operator K_b are defined by their actions on an arbitrary orbital $\chi_a(1)$:

$$J_b(1)\chi_a(1) = \left[\int dx_2 \chi_b^*(2) \frac{1}{r_{12}} \chi_b(2) \right] \chi_a(1) \quad (14)$$

$$K_b(1)\chi_a(1) = \left[\int dx_2 \chi_b^*(2) \frac{1}{r_{12}} \chi_a(2) \right] \chi_b(1) \quad (15)$$

By performing a unitary transformation on the occupied orbitals, we diagonalize the matrix ϵ_{ba} , reducing the coupled equations to the canonical form, this is shown in detail in [1] but we will skip the full derivation here:

$$f(1)\chi_a(1) = \epsilon_a \chi_a(1) \quad (16)$$

Eq.(16) is in terms of the spin orbitals these consist of a spatial and a spin component. The operators have no actual dependence on spin so this part of the integral will cancel and we are left with just spatial wavefunctions [6] as seen:

$$f(1)\psi_a(1) = \epsilon_a \psi_a(1) \quad (17)$$

This equation or set of equations will reappear constantly and will be the focus of our study, we will aim to solve the equations to obtain energy values for our electron orbitals and use these to calculate the energy of the atom to compare with other studies and experimental results.

When calculating the total energies it would be easy to assume that this is just equal to the sum of all of the orbitals, however it is important to note that this would result in the double counting of all of the electron electron interactions between pairs, so we must account for this. The total energy of the atom can be calculated in Eq.(11). This can be simplified as it is possible to reach the final energy with the addition of the core energy values to the total of the orbitals and multiplying this sum by 0.5. This average solution is neat and easy:

$$E_{total} = \frac{1}{2} \sum (\epsilon_i + \langle \psi_i | H_{core} | \psi_i \rangle) \quad (18)$$

where $H_{core} = \mathbf{T} + \mathbf{V}_{nuc}$.

2.4 The density matrix + the coulomb and exchange operators

To calculate the Coulomb ($\hat{J}$) and Exchange ($\hat{K}$) operators for our Fock matrix, we first need a mathematical map of the electron distribution. The probability of finding an electron in a specific region of space is given by the total electron density, $\rho(\mathbf{r})$, which is the sum of the squared modulus of the occupied molecular wavefunctions.

For a closed-shell system with N electrons (where each spatial orbital is doubly occupied), we sum over the $N/2$ occupied spatial orbitals. By expanding these unknown molecular orbitals (ψ_a) into our known basis functions (ϕ_μ), we can extract the density matrix, $P_{\mu\nu}$:

$$\begin{aligned} \rho(\mathbf{r}) &= 2 \sum_a^{N/2} \psi_a^*(\mathbf{r}) \psi_a(\mathbf{r}) \\ &= 2 \sum_a^{N/2} \sum_\nu C_{\nu a}^* \phi_\nu^*(\mathbf{r}) \sum_\mu C_{\mu a} \phi_\mu(\mathbf{r}) \\ &= \sum_{\mu\nu} \left[2 \sum_a^{N/2} C_{\mu a} C_{\nu a}^* \right] \phi_\mu(\mathbf{r}) \phi_\nu^*(\mathbf{r}) \\ &= \sum_{\mu\nu} P_{\mu\nu} \phi_\mu(\mathbf{r}) \phi_\nu^*(\mathbf{r}) \end{aligned} \quad (19)$$

The density matrix, $P_{\mu\nu}$, is the discrete, computational representation of the electron density [1]. It is the crucial link in our iterative process. In the Hartree-Fock approximation, a single electron is modeled as feeling the repulsion of all other electrons not as individual moving point charges, but as an average, static field. However, to calculate this average field (the potential), we need to know the wavefunctions of the electrons. Yet, to find the wavefunctions, we must first solve the Schrödinger equation, which requires the potential. This paradox is resolved computationally via the Self-Consistent Field (SCF) method [5]. Through the use of initial guesses for the wavefunctions, we can build an initial density matrix. This matrix is then used to construct the $\hat{J}$ and $\hat{K}$ operators, which form the Fock matrix to be diagonalized, yielding a new, improved density matrix. This is one of the main approximations we make as we need more variables than we have at a given time to calculate our energy but through iteration we can obtain accurate solutions nonetheless.

With the density matrix defined, the next computational challenge is constructing the specific expressions for the Coulomb ($\hat{J}$) and Exchange ($\hat{K}$) operators. Transitioning from continuous equations to numerical approximations significantly simplifies the problem: our continuous spatial wavefunctions are evaluated on a 1D grid to become vectors, meaning our quantum mechanical operators can be represented as matrices.

The Coulomb operator represents the classical electrostatic repulsion. The potential created by an electron in orbital j acting on an electron in orbital i is defined as:

$$\hat{J}_j(r_1)\psi_i(r_1) = \left[\int \frac{|\psi_j(r_2)|^2}{r_{12}} dr_2 \right] \psi_i(r_1) \quad (20)$$

By summing over all occupied orbitals, the total Coulomb potential simply relies on the total electron density (ρ). Because this is a local operator—the potential at r_1 only depends on the surrounding charge distribution—computationally, it evaluates to a 1D vector. To incorporate this into the Fock matrix, this vector is placed along the diagonal of the J matrix, with all off-diagonal elements set to zero.

The Exchange operator, however, is purely a quantum mechanical consequence of the Pauli exclusion principle making it more difficult. Based on the derivation by Szabo and Ostlund, its action on a wavefunction involves swapping the coordinates of the two electrons:

$$\hat{K}_j(r_1)\psi_i(r_1) = \left[\int \frac{\psi_j^*(r_2)\psi_i(r_2)}{r_{12}} dr_2 \right] \psi_j(r_1) \quad (21)$$

By factoring out $\psi_i(r_2)$ from the integral and summing over all occupied orbitals j , we can express the total exchange operator in terms of the non-local density matrix, $P(r_1, r_2) = \sum_j \psi_j(r_1)\psi_j^*(r_2)$:

$$\hat{K}(r_1, r_2)\psi_i(r_1) = \int \left[P(r_1, r_2) \frac{1}{r_{12}} \right] \psi_i(r_2) dr_2 \quad (22)$$

Because the operator involves evaluating the wavefunction at two different coordinates simultaneously, it is non-local. In our computational program, this means the K matrix cannot be diagonalized into a single vector like J . Instead, it forms a full matrix where the elements are computed as the element-wise product of the density matrix and the $1/r_{12}$ interaction kernel.

The critical term linking both of these operator calculations is $1/r_{12}$. Read on to find out how this specific interaction is evaluated.

2.5 Multipole expansion theorem - shapes of the atomic orbitals

The Coulomb and Exchange operators fundamentally rely on evaluating the electron-electron repulsion term, $1/r_{12}$. To compute this within the framework of a 1D radial grid, this term must be decoupled into its radial and angular components using the Laplace multipole expansion [7]:

$$\frac{1}{r_{12}} = \sum_{l=0}^{\infty} \sum_{m=-l}^{+l} \frac{4\pi}{2l+1} \frac{(r_{<})^l}{(r_{>})^{l+1}} \times Y_{lm}^*(\theta_1, \phi_1) Y_{lm}(\theta_2, \phi_2) \quad (23)$$

For the first two atomic structures of interest, Lithium and Beryllium, the calculations only involve s-orbitals. Because the electron density cloud of an s-orbitals is spherical, $l = 0$. This simplifies the mathematics significantly; the spherical harmonics equal a constant $1/\sqrt{4\pi}$, which cancels out the 4π in the expansion, removing all angular dependence. The radial terms reduce to $(r_{<})^0/(r_{>})^1$, leaving simply $1/r_{>}$, where $r_{>}$ represents the greater of the two radial coordinates being compared. Computationally, this allows the interaction kernel to be built in a single line using a maximum value function.

However, for higher values of l , such as $l = 1$ in the case of Boron's $2p$ electron, the system is no longer spherical. Expanding the $l = 1$ term explicitly yields the dipole interaction[8]:

$$\text{Dipole Term} = \frac{r_{<}}{r_{>}^2} \left(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(\phi_1 - \phi_2) \right) \quad (24)$$

The use of this angular dependence will in all cases be integrated through the calculation for $\mathbf{K}$ matrices. The $\mathbf{J}$ matrix is calculated with a spherical approximation. As nothing else depends on this result it is possible to calculate separately the effect of the angular difference between a dipole and a sphere and just substitute the coefficient into the final equation.

The full term we have for the spherical harmonic dependence and sum over $m = \pm l$ without the radial component is :

$$\begin{aligned} \text{Angular dependence} &= Y_{1,-1}^*(\theta_1, \phi_1) Y_{1,-1}(\theta_2, \phi_2) \\ &+ Y_{1,0}^*(\theta_1, \phi_1) Y_{1,0}(\theta_2, \phi_2) \\ &+ Y_{1,1}^*(\theta_1, \phi_1) Y_{1,1}(\theta_2, \phi_2) \end{aligned} \quad (25)$$

This we need to integrate with respect to a sphere and dipole spherical harmonics choosing just one axis for the dipole we can see that the integral will cancel out to the result of: $1/3$ with this formula:

$$\int Y_{L_1, m_1}(\theta, \phi) Y_{L_2, m_2}(\theta, \phi) d\Omega = \delta_{L_1, L_2} \delta_{m_1, m_2} \quad (26)$$

We can perform this integral:

$$\begin{aligned} &\frac{4\pi}{3} \int Y_{0,0}^*(\theta, \phi) Y_{1,0}(\theta, \phi) \\ &\times [\text{interaction}] \\ &\times Y_{0,0}(\theta, \phi) Y_{1,0}^* d\Omega \end{aligned} \quad (27)$$

This integral produces our $1/3$ term to use for calculation of the exchange term between spherical and dipole shapes.

While the multipole expansion extends to infinity, calculating just the first non-vanishing term is sufficient for the ground state of Boron. Higher-order terms (like $l = 2$) would only become strictly necessary if we were calculating the interactions of d -orbitals in transition metals.

2.6 Restricted and Unrestricted Hartree Fock models - Koopmann's theorem

Having established the formalism for constructing the Fock matrix, we now distinguish between the restricted and unrestricted Hartree–Fock (HF) methods, and their applicability to closed- and open-shell systems [1].

In the Restricted Hartree–Fock (RHF) method, each spatial orbital is doubly occupied by electrons of opposite spin. Consequently, the spin-up (α) and spin-down (β) electrons share the same spatial orbital, and the total electron density can be described using a single density matrix. This approach is therefore appropriate for closed-shell systems, such as Beryllium which was the first atom we looked at having 2 electron pairs in each shell.

For such systems, the Fock matrix is constructed as

$$F = T + V + J - K, \quad (28)$$

Starting from an initial guess for the orbitals, $1s$ and $2s$, one constructs the density matrix, from which the Coulomb and Exchange matrices are evaluated. Solving the resulting Fock matrix eigenvalue problem yields improved approximations to the orbital energies, with the lowest eigenvalues corresponding to the occupied orbitals.

Within this framework, Koopmans' theorem provides an approximate interpretation of calculated orbital energies. It states that the first ionisation energy is approximately equal to the negative energy of the highest occupied molecular orbital [1]. This theorem assumes the remaining electrons do not 'relax' or rearrange upon the removal of the electron. We will use this theory to validate our results.

While RHF is efficient and well-suited to closed-shell atoms, it becomes inadequate for open-shell systems. In such cases, the unrestricted Hartree–Fock (UHF) method is preferred for example with Lithium. Here if one was to use the same method then an exchange term between an α spin and β spin electron would be calculated however this interaction does not exist. The UHF method by Pople-Nesbet from the core textbook [1] follows the same process for derivation as we have seen, but uses a combination of different density matrices to represent each spin and in some cases later each orbital shape. This leads to two coupled Fock equations with a similar derivation to earlier:

$$F^\alpha = T + V + J[P^\alpha + P^\beta] - K[P^\alpha], \quad (29)$$

$$F^\beta = T + V + J[P^\alpha + P^\beta] - K[P^\beta]. \quad (30)$$

This correctly reflects the physical fact that exchange interactions do not occur between electrons of opposite spin. The equations may be solved separately because the exchange interaction only occurs between electrons of the same spin, allowing the total Fock matrix to decouple into independent α and β equations that share only the total electrostatic potential.

For atoms such as Lithium or Boron, which contain unpaired electrons, this results in separate equations for each spin set of electrons, allowing greater flexibility in the wavefunctions and improved accuracy compared to RHF. Furthermore, with Boron it requires a density matrix for the $\mathbf{p}$ -orbital due to a different interaction coefficient with the shape of Boron's $\mathbf{p}$ -orbitals no longer being spherical.

For higher-order orbitals, such as those encountered beyond the ground-state of Beryllium, the assumption of spherical symmetry is no longer strictly valid. In particular, p -orbitals possess angular dependence. Since the present simulation framework relies on a spherically symmetric potential, an additional approximation is required.

To maintain spherical symmetry, the occupation of a single $\mathbf{p}$ -orbital (e.g., p_x) is replaced by an equal distribution over all three degenerate p -orbitals. That is, the electron density is averaged such that each of p_x , p_y , and p_z carries one-third of the occupation. This procedure effectively restores a spherically symmetric charge distribution, allowing the use of the central field approximation.

However, this averaging introduces additional contributions to the electron–electron interaction terms. In particular, when considering interactions between $\mathbf{s}$ and $\mathbf{p}$ -orbitals, it becomes necessary to include dipole-like corrections arising from the non-spherical character of the underlying orbitals as seen in section 2.5. These terms ensure that the interaction between the averaged $\mathbf{p}$ -electron density and the $\mathbf{s}$ -orbitals is treated consistently within the approximation.

2.7 Note on orthogonality and the spectral theory.

In the Hartree Fock model it is essential that the resulting orbitals form an orthonormal set in both the restricted and unrestricted case. In many standard quantum chemistry implementations, the use of non-orthogonal basis sets (such as Gaussian-type orbitals) leads to a generalised eigenvalue problem of the form $[FC = SCE]$ as we saw earlier in the derivation, where S is the overlap matrix. In such cases, an explicit orthogonalisation procedure is required. However, as we implement a finite difference grid, this corresponds to a discretised real-space basis which is inherently orthogonal, such that the overlap matrix reduces to the identity ($S = I$). As a result, the eigenvalue problem simplifies to $[FC = CE]$ and the resulting eigenfunctions are automatically orthonormal.

3 Methods and code

3.1 Initial set up and functions

Having established the underlying theory, we now examine the practical implementation of the code. The primary objective is to iteratively construct the Fock matrix, using it to refine our wavefunction estimates until the system's total energy reaches a minimum, self-consistent value.

The numerical framework begins with a finite difference grid. This is an array of N points spaced by a small interval dr . To avoid numerical instabilities and the $1/r$ or $1/r^2$ singularity at the nucleus, the grid starts at $r = dr$ rather than zero. The grid size is tailored to the expected atomic orbital radius for a given atomic number Z , which defines the number of protons for the atom of interest. We must also provide the

initial guesses for the wavefunction that will be updated iteratively, the reduced hydrogenic wavefunctions were first trialed, the aim is to have the solutions to converge quite quickly so the initial wavefunction choice is something that will be explored.

The Hamiltonian components from Eq.(1) are constructed in matrix form. The kinetic energy matrix $\mathbf{T}$ is implemented using the central finite difference method discussed. This being diagonal of -2 and off-diagonals of 1, as shown in the appendix. The nuclear potential matrix V_{nuc} is a diagonal matrix with elements $-Z/r$. The Hamiltonian components from Eq.(1) are constructed in matrix form.

The electron-electron interaction requires the construction of the r_{12} multipole kernel. For the monopole interaction, the kernel is constructed using the 'outer product' and 'maximum' functions to efficiently determine $r_{>}$ without nested loops. For systems requiring dipole terms Eq.(23), a separate kernel is maintained to handle the specific $\frac{r_{<}}{r_{>}}$ dependence. The angular components are handled separately within the exchange $\mathbf{K}$ matrix. See the appendix for the specific cases.

The density matrices $\mathbf{P}$ are computed from the current iterations wavefunctions as seen in Eq.(19). The coulomb matrix $\mathbf{J}$ is formed from the charge density which is the diagonal of $\mathbf{P}$ multiplied by the kernel and integrated Eq.(20). The exchange matrix $\mathbf{K}$ is calculated by through multiplication of $\mathbf{P}$ and the kernel integrated Eq.(21).

Differences for the restricted vs the unrestricted method; For the unrestricted method we must use multiple density matrices, this is because it is easier to separate and stop an interaction term being created between an α and β spin set which does not exist. This is simply achieved by maintaining separation for each spin, multiple density matrices and Fock equations. Furthermore when we approach higher number of electron elements such as Boron we need to consider the dipole term in the multipole expansion so we have another function for the r_{12} value, and the angular momentum term from Eq.(3). Therefore, we have another density matrix for the $\mathbf{p}$ -spin state and Fock equation. The second value of r_{12} also needs an accompanying $\mathbf{K}$ function for itself and the factor 1/3 mentioned in section 1.5.

3.2 The Self consistent field loop

With the functional components established, the system is solved using the Self-Consistent Field (SCF) method [5]. The SCF procedure is an iterative algorithm designed to find the set of orbitals that minimise the total energy functional by solving the effective one-electron equations repeatedly until the field "seen" by the electrons becomes stationary. The SCF loop iteratively solves the Fock equations for the energy levels, until the change in orbital energy falls below a specified tolerance. The loop created has a maximum number of iterations in-order to stop the code for cases where the wavefunctions do not converge. There is a limit set for how strict we want the convergence. Once convergence is achieved, the final result is used to calculate the total ground-state energy.

Beryllium: For the case of the restricted method with Beryllium it is assumed both electrons in the 1s and the 2s to have the same wavefunction. A single Fock matrix is then constructed from the initial wavefunctions with the functions above Eq.(17). This is solved for the 1s and 2s states. The

program does all the heavy lifting for us with:

$$\text{Eigenvalues, Eigenvectors} = \text{eigsh}(\text{Fock}, k = 2, \text{which} = 'SA') \quad (31)$$

This finds the 'smallest algebraic' eigenvectors and eigenvalues which correspond to our new wavefunctions and the energy levels of the orbitals respectively. Then we check our convergence and if the limit has been reached the program can then calculate the total energy of the atom from the individual electron energy. As mentioned earlier this is not simply the sum of all of the individual electron energies as there will be aspects of the problem which will be doubly counted. The energy is calculated with Eq.(18).

Lithium: The unrestricted method allows us to have different energies for the electrons in each shell which is a more accurate approximation as the electrons on the same orbital behave differently. This is done by having two density matrices, one for each spin value, P_α and P_β . We make these from our three initial guesses, one for each electron. The first consists as you would expect of the $1s\alpha + 2s\alpha$ and the second from the $1s\beta$. These will be used similarly to Beryllium to calculate the $\mathbf{J}$ and $\mathbf{K}$ values, and in-turn to create the Two Fock matrices that we will solve as seen in Eq.(29) & Eq.(30). The $\mathbf{J}$ matrix uses a total density matrix because it is using just the diagonal terms this is acceptable so it uses $\mathbf{P} = P_\alpha + P_\beta$, and calculates as Beryllium does - first calculating the charge density ρ and we have one value of $\mathbf{J}$ coulomb interaction as per Eq.(20). The difference comes from calculating two different exchange $\mathbf{K}$ matrices, which calculates one from each spin set to ensure we calculate the exchange between electrons with the same spin. There still needs to be this calculation for the $1s\beta$ electron as the $\mathbf{K}$ term removes the doubly counted self exchange term in $\mathbf{J}$. Now it is possible to build the two Fock matrices and solve them separately as we did for the single in Beryllium. This again produces the new wavefunction where we continue the process until convergence, giving us the final wavefunctions and energies of our electrons.

Boron: Boron also uses the unrestricted method however the code becomes more complicated due to the dipole shape of the $\mathbf{p}$ -orbital. The same principle is used, in this case we use a third density matrix for the $\mathbf{p}$ -orbital with spin α , this will become clear shortly. The same way we totaled the density matrices for the $\mathbf{J}$ matrix calculation in Lithium we can repeat the process here for Boron. This time we have to calculate 3, $\mathbf{K}$ matrices for 3, Fock equations - this is due to the fact that our $l = 1$ so we have the centrifugal term for the new orbitals seen in Eq.(3). We calculate the different $\mathbf{K}$ matrices dependent on what the electrons can 'see'. Simply for the β electrons have the interaction between the two $\mathbf{s}$ -orbitals which uses the monopole term and P_β density matrix. The α electrons in $\mathbf{s}$ -orbitals feel two interactions, the interactions between 1s and 2s with the monopole term and the interaction of them both with the P_α orbital, this we have to use the dipole like term in the calculation with the $\mathbf{p}$ -orbital density matrix. Then finally the $\mathbf{p}$ -electron will feel an interaction potential from the 1s and 2s electrons calculated with the P_α density matrix and the dipole interaction term, and not to forget to remove the self interaction. Finally we can construct the 3 Fock equations:

$$F_{sa} = [T + V_{nuc} + J_{mat} - K_{mat_{sa}}] \quad (32)$$

$$F_{sb} = [T + V_{nuc} + J_{mat} - K_{mat_{sb}}] \quad (33)$$

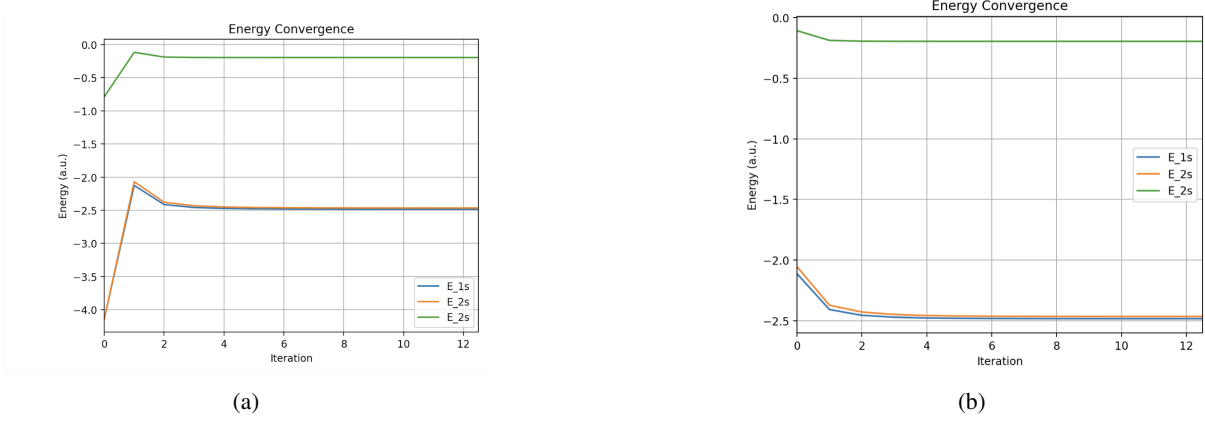


Figure 2: Comparison between the convergence of a) r , and b) Hydrogenic functions.

$$F_{sp} = [T + V_{nuc} + V_{centrifugal} + J_{mat} - K_{mat_{sp}}] \quad (34)$$

Then the process is the same as above, solving for the final wavefunctions and energies, except for the final energy calculation, it is essential to include the centrifugal term in the core energy Hamiltonian calculation.

Carbon: The extension then to Carbon is relatively simple, we are working with another electron with α spin, due to Hund's rule of shell filling, most things remain unchanged. The electron density is kept as 1 electron as it is the method the code structure allows for. We must multiply the interaction term for s -orbitals with p -orbitals by 2, and there is an additional p -orbital exchange term as we now have two α electrons in the sub-shell. This has a predetermined angular dependence of $2/25$. This is the dipole-dipole interaction term and can be derived similarly to the $1/3$ in Eq.(27).

A note about the cancellation term we have mentioned using Lithium as our example. When we construct the coulomb term it is an average field of the three electrons and then we see how each electron interacts with the field, thus creates an exchange between an electron and itself. Through the $\mathbf{K}$ matrix we cancel out this unnecessary term.

4 Results and discussion

Having established the required theoretical framework and the numerical method we can now evaluate the results and performance of the code. This section will first discuss the different options for initial conditions and the convergence of these, and then discuss the results of the atomic energy levels.

4.1 Optimisation of initial conditions.

To determine the most effective conditions of the radial grid and efficient initial wavefunctions to use for the project, the test case of Lithium was used. The literature value for the Hartree Fock energy of Lithium is $-7.4332E_h$. We found that a radial grid defined with $N = 800$ points and a maximum radius of $12.5a.u.$ was sufficient to achieve results close to this value, of $-7.4285E_h$. The relationship between grid density and accuracy is non-linear. While increasing the point density, e.g to $N = 1000$, only marginally improves the resolution of the region near the nucleus we are most interested in so the result only improves slightly. The choice of $N=800$ captures the $1s$ electrons and the tail of the $2s$ orbital with sufficient

detail and provides a balance between numerical precision and computational efficiency. The program requires initial guesses for the wavefunctions. The different guesses were, the actual grid itself, the r value at each point and, the reduced radial wave functions. For Lithium these are:

$$u_{1s}(r) = 2Z^{3/2}re^{-Zr} \quad (35)$$

$$u_{2s}(r) = \left(\frac{Z}{2}\right)^{3/2} r \left(1 - \frac{Zr}{2}\right) e^{-Zr/2} \quad (36)$$

Both initial conditions converged to the same numerical value of $-7.4285E_h$, however the rate at which they did differ.

Whilst using the initial conditions of the reduced radial wavefunctions the convergence rate is faster at 20 iterations to that of 22 for the simple r function. The difference is minimal which is interesting and shows the 'strength' of the underlying physical properties. It is expected that the reduced radial wave functions will converge faster as they have a built in similarity in shape. Therefore, the radial wavefunctions were used for the subsequent atoms in the report to ensure that as the complexity of the problem increases there is an initial backbone to our function present.

4.2 Energy levels

The results presented in table (1) show the total energy comparison and demonstrate that the numerical model is consistent. While the accuracy of the total Hartree-Fock energy is limited by the radial grid density, a resolution of $N=800$ points over the defined radial distance provides a sufficient balance between computational cost and physical detail. The deviations from exact experimental values are primarily due to the approximations of the Hartree-Fock method, the model treats the electrons as an average moving energy which neglects the correlation energy. Our grid also starts at $r = dr$ and as we reach the larger atoms we split an electron across orbitals.

4.3 Pauli exclusion and exchange splitting

A key validation of the code is that it demonstrates the Pauli exclusion principle, the difference between the energy of electrons on the same orbit but having different spin have different bound energy see table (2). For instance in the Lithium system the $1s\alpha$ electron with up spin is more tightly bound (more

Table 1: The total Hartree Fock energies of atoms, compared to another set of numerical values and the exact energies. The values are from Hartree-Fock Values Of Energies [9].

Atom	Elec config	E_{total}	$E_{reference}$	$\Delta_{reference}$	E_{exact}
Li	$1s^2 2s^1$	-7.4285	-7.4332	-0.0047	-7.4786
Be (Rhf)	$1s^2 2s^2$	-14.5673	-14.5752	-0.0079	-14.6693
Be (Uhf)	$1s^2 2s^2$	-14.5673	-14.5752	-0.0079	-14.6693
B	$1s^2 2s^2 2p^1$	-24.4857	-24.5351	-0.0494	-24.6591
C	$1s^2 2s^2 2p^2$	-37.5734	-37.7022	-0.1288	-37.8574
$C_{p_y \downarrow}$	$1s^2 2s^2 2p^2$	-37.8075	-37.7022	–	-37.8574

Table 2: Each orbital compared to the reference values from by Bunge et al. (1993) [10].

Atom	Elec config	$1s \uparrow$	$1s \downarrow$	$2s \uparrow$	$2s \downarrow$	$2p_x \uparrow$	$2p_y \uparrow$
Li	$1s^2 2s^1$	-2.4837	-2.4656	-0.1961	–	–	–
Li Reference	$1s^2 2s^1$	-2.5233	-2.4764	-0.1963	–	–	–
Be Restricted	$1s^2 2s^2$	-4.7292	-4.7292	-0.3088	-0.3088	–	–
Be Reference	$1s^2 2s^2$	-4.7327	-4.7327	-0.3093	-0.3093	–	–
B	$1s^2 2s^2 2p^1$	-7.6769	-7.6619	-0.5414	-0.4436	-0.3102	–
B Reference	$1s^2 2s^2 2p^1$	-7.7663	-7.6493	-0.5694	-0.4431	-0.3091	–
C	$1s^2 2s^2 2p^2$	-11.3049	-11.2610	-0.8280	-0.5867	-0.4071	-0.4071
C Reference	$1s^2 2s^2 2p^2$	-11.3934	-11.3130	-0.7816	-0.6273	-0.4333	-0.4333

negative) than its $1s\beta$ spin down counterpart. This energy gap is a direct consequence of the exchange term in the UHF equations. By reducing the energy of parallel spin pairs the model is successful in demonstrating the exchange principle. Similar effects can be seen from the results of the other atoms. The electrons in Carbon follow the trend with a scaling factor, larger gaps appear as there is a larger ratio of electrons with spin up compared with down, the effects seen are greater.

The last row in table (1) shows an example of Hund’s rule. By comparing a configuration where the 2p electrons have opposite spins α, β compared to the ground state α spins, we can see evidence for the exchange term. The parallel spin case has a lower energy as we would expect due to the separation of the electrons average position meaning there is less repulsion and there are more bound.

4.4 Orbital energy

Table (2) shows the energy of all of the orbitals in our system, close agreement between our calculated energy orbitals and established literature values [10] serves as a verification of the accuracy of our Hartree-Fock implementation. We can see that the model agrees with the principle of 1s orbitals having the most negative energies, followed by 2s and 2p.

Comparison between the RHF and UHF methods for Beryllium table (2) shows identical total energies and orbital

eigenvalues. Despite introducing variations in the initial trial wavefunctions, the UHF solver naturally collapses to the RHF solution. This is consistent with the findings of Szabo and Ostlund [1], as Beryllium is a ”closed-shell” system. With the system having equal spin states, the symmetry of the Hamiltonian ensures that the α and β potentials remain identical.

4.5 Radial Probability densities

The radial probability density can be extracted from our results, defined as $P(r) = |U(r)|^2$, it provides a visual representation of electron position as a distance from the nucleus. Comparing Lithium ($Z=3$) and Boron ($Z=5$) highlights the effect of increasing nuclear potential on atomic structure.

As seen in Figure [3] the electrons in Boron are much more tightly bound to the nucleus due to the stronger coulomb attraction. The 1s electrons in Boron are twice as close than that of Lithium. The initial small peak of the 2s electrons is also larger in the Boron atom, this is because the 1s electron is drawn into the nucleus and the 2s electron can follow.

Figure [3] also shows the 2s wavefunction has two peaks, one of which is very close to the nucleus, within the distribution for the 1s electron. This allows the 2s electron to penetrate the shielding provided by the 1s electrons and experience a significantly higher effective nuclear charge. In contrast, the 2p orbital lacks an inner node and is largely excluded from

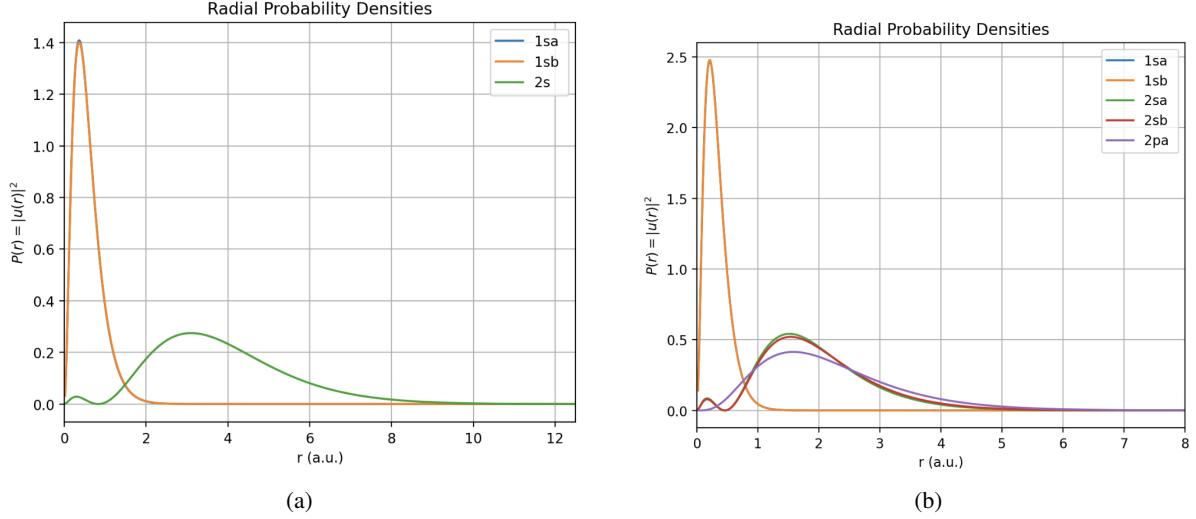


Figure 3: Comparison between the radial probability distributions of a) Lithium, and b) Boron.

the core region, meaning it is more effectively shielded and resides in a region of lower effective nuclear attraction. This relationship is quantified by the expression:

$$E_n \approx -13.6, \text{eV} \frac{Z_{eff}^2}{n^2} \quad (37)$$

Because the 2s electron experiences a larger Z_{eff} due to this penetration, its energy level is pulled lower (more bound) than the 2p electron, even though they share the same principal quantum number $n=2$, this agrees with our results.

The effects of the Pauli exclusion principle can also be seen in the radial distribution. Looking at Boron the 2s electrons have a slight splitting and it can be seen that the α electron is experiencing a shift closer to the nucleus whereas the β electron does not feel the 'push' from the p -orbitals as there is no exchange between them.

Figure [4] is a plot of the radial probability densities of the wavefunctions we have found for Boron along side that of the hydrogenic reduced radial wavefunctions representing the model of the independent particle model. The independent particle model essentially neglects the electron-electron repulsion terms and deals only with the analytically solvable Hamiltonian. While the 1s curves are nearly identical due to the unshielded nuclear charge at short distances, the 2s HF density is pushed outward relative to the IPM. This is due to the

screening effect of the 1s electrons, meaning the 2s electrons feel a reduced effective Z .

The 2p density, however, appears more contracted than the hydrogenic 2p model. This occurs because the IPM 2p wavefunction is calculated for a single electron around a nucleus. In the HF model, the 2p electron "sees" a nucleus that is being shielded by 1s and 2s electrons, but the Exchange interaction with the $1s\alpha$, $2s\alpha$ means that they are found on average further from the $2p\alpha$ electron, thus in turn reduces the coulomb interaction.

4.6 Koopmans theorem

To further validate the accuracy of the model we can apply Koopman's Theorem. Mentioned previously in the theory section, it approximates the first ionisation energy for an atom as the energy level of the highest occupied individual energy level by assuming the other orbitals do not 'relax'. The results indicate that Koopman's Theorem is a reliable approximation for simpler systems like Lithium, Boron and Carbon. However, with the full shell model of Beryllium we have a slight disparity. For a reference the first ionisation energies of the respective atoms are: Lithium = 0.1981, Beryllium = 0.3426, Boron = 0.3049 and Carbon = 0.4138. The approximation doesn't hold well when dealing with a full shell but our results

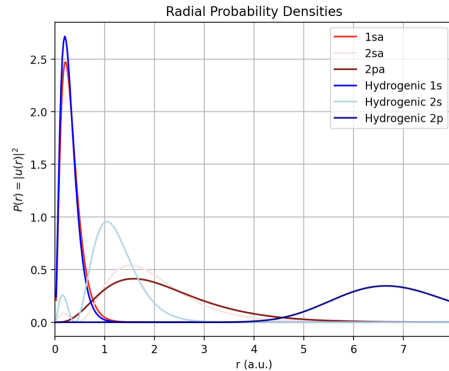


Figure 4: Comparison for IPM against our radial functions.

are further validated here.

5 Further development

The success of the current model for lower values of Z for producing the ground state energies of atoms and radial distributions suggests that the model could be extended to larger atomic systems, there are several refinements and extensions that would need to be considered to keep the model accurate as we can already see a slight reduction of accuracy with the model for Carbon due to the complex nature of the interaction between dipoles. The code should follow the same procedure and increase the $\mathbf{K}$ matrix calculations, when we progress to orbitals beyond the $\mathbf{p}$ shape a different multipole expansion term will have to be calculated and used in the exchange term, for instance the $\mathbf{D}$ orbitals will require a quadrupole term. This will increase the complexity of the $\mathbf{K}$ matrix calculation for the potential each electron 'sees'. It is essential we still follow Hund's rule of electron filling, starting with parallel spin states before the anti-parallel orbitals as this gives the lowest available energies.

In theory we can keep using the splitting of electron model but as we increase the size of the atom I am not sure the validity of the results will remain the same as we do begin to lose accuracy in the total Hartree energy for Carbon. As we increase the size of the atom it is crucial that the relativistic effects of the atom are also considered. When increasing the size of the atom the relativistic effects become unavoidable, as we know hydrogen even has slight relativistic behaviour. When approximating for smaller atoms we will be close to the actual result, it is only becoming noticeable as we approach larger Z values. Then we will need to use the Dirac equation and a different approximation.

The code could be improved further with a logarithmic grid, this in short fills the orbitals more effectively with points where they are more useful, it tightens the spread to surround the nucleus where the main interactions are taking place so is more accurate. However this requires then the orthonormalisation of the wavefunctions which we did not have to consider with the standard grid used.

6 Conclusion

The primary objective of this project and report was to develop a computational method and provide a background to the Hartree Fock method, building on a topic from the PHYS261 course. The aim was to create a project that would be able to be used as a resource and follow similar methods for coding as that of the course. By using a finite-difference radial grid and a matrix-based approach, the project successfully transitioned from theory to a functional computational tool.

The results from the project are promising, the calculated energy levels and radial distribution agrees with literature and reproduces key physical phenomena.

While the model demonstrates high precision for light atoms, the analysis of heavier systems like Carbon highlights the inherent limits of non-relativistic, mean-field approximations. It is my hope that this report and the accompanying code can serve as a practical resource for future students of PHYS261, there is still a lot more development that can be done with this theory and improvements to the code.

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A.i

How did I use A.i to aid with the project?

- Initial source searches are quite easy to pinpoint with A.i so I obtained a couple of initial references with prompts such as 'Hartree Fock theory', or 'Hartree Fock code'.
- Code: small fixes and errors are easily spotted, improved code for efficiency when trying to achieve better accuracy the run times were long so I asked Gemini to alter a couple of the main lagging components for instance the **J** and **K** matrix integration, and some if statements A.i. built quicker components for these to improve run times of the integration.

- Code: Within the multipole expansion I had the formula from Morten's notes for the 261 course but did not know how to integrate over the angular component, Gemini provided a target value of 1/3 the integral was over spherical harmonics which I could then calculate the analytical solution for and implement into my code knowing the answer being 1/3 and this improved the Boron value, similarly the 2/25 term for the p-p exchange in carbon could then be calculated.
- Code: **K** matrix building - I did not realise that we were able to drop the other wave function from the calculation of the **K** matrix, A.i pointed out here that as we are using linear algebra a simple matrix with out the term could be used and when we diaganolise with the guesses it would correct itself through the iterations of the SCF, making the problem much simpler.
- Code: A.i reminded me to include the self interaction of the single electric orbitals - a simple fix to improve accuracy.
- The import of the code to the appendix into LaTeX format.
- Report : Spell check was done and slight improvement of writing but kept as most of my own wording.

Appendix: Hartree-Fock Implementation for Lithium

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.sparse import diags
4 from scipy.sparse.linalg import eigsh
5
6 # --- Parameters ---
7 Z = 3
8 N = 750
9 R = 12.5
10 dr = R / (N + 1)
11 r = dr * np.arange(1, N + 1)
12 # Setting up a radial grid which starts at dr rather than 0 as to avoid singularity
13
14 # --- Functions ---
15
16 def build_T_operator(N, dr):
17     main = np.full(N, -2.0)
18     off = np.ones(N - 1)
19     D2 = diags([off, main, off], offsets=[-1, 0, 1], shape=(N, N)) / (dr**2)
20     return (-0.5) * D2
21 # This is our kinetic energy operator, creates a matrix N by N of -2, 1 and 0's
22
23 def normalize_u(u, r):
24     norm = np.sqrt(np.trapezoid(u**2, r))
25     return u / norm
26 # Takes in the wavefunction and returns a normalised version
27
28 def build_r12_matrix(r):
29     return 1.0 / np.maximum.outer(r, r)
30 #Building the kernal, checks which value of rij is greater
31
32 def build_p_a_matrix(u1, u2):

```

```

33     return np.outer(u1, u1) + np.outer(u2, u2)
34 # Density matrix for Alpha spin electrons - we will have 2 in the case of lithium
35
36 def build_p_b_matrix(u1):
37     return np.outer(u1, u1) # One s down electron
38 # Density matrix for Beta eletron
39
40 def calculate_J(P, r12_kernel, dr):
41     rho = np.diag(P)
42     J_vec = np.dot(r12_kernel, rho) * dr
43     return diags(J_vec, 0)
44 # Calaculation input is the total density matrix and returns J matrix
45
46 def calculate_K_UHF(P, r12_kernel, dr):
47     return P * r12_kernel * dr
48 # K matrix is built from this function
49
50 # --- SCF Setup ---
51 T = build_T_operator(N, dr)
52 V_nuc = diags(-Z / r, 0)
53 kernel = build_r12_matrix(r)
54
55 # Initial Guess (Hydrogenic)
56 u1sa = normalize_u(r*(Z)**1.5 * 2 * np.exp(-Z * r), r)
57 u1sb = normalize_u(r*(Z)**1.5 * 2 * np.exp(-Z * r), r)
58 u2sa = normalize_u(r*(Z/2)**1.5 * (1 - Z*r/2) * np.exp(-Z*r/2), r)
59
60 MAX_ITER = 50
61 TOLERANCE = 1e-7
62 E_1sa_history, E_1sb_history, E_2sa_history = ([ ] for _ in range(3))
63
64 # --- Main SCF Loop ---
65 for i in range(MAX_ITER):
66     Pa = build_p_a_matrix(u1sa, u2sa)
67     Pb = build_p_b_matrix(u1sb)
68     P = Pa + Pb
69
70     J_mat = calculate_J(P, kernel, dr)
71     K_mat_a = calculate_K_UHF(Pa, kernel, dr)
72     K_mat_b = calculate_K_UHF(Pb, kernel, dr)
73
74 # Building the matrixes components we need to form the fock matrices below
75
76     Fock_a = T + V_nuc + J_mat - K_mat_a
77     eigenvalues_a, eigenvectors_a = eigsh(Fock_a, k=2, which='SA')
78
79     Fock_b = T + V_nuc + J_mat - K_mat_b
80     eigenvalues_b, eigenvectors_b = eigsh(Fock_b, k=1, which='SA')
81
82 # The eigsh function solve the code for the smallest algebraic values for the
83     eigenvectors and eigenvalues these are orthogonal even accross the two matrices.
84
85     new_u1sa = normalize_u(eigenvectors_a[:, 0], r)
86     new_u2sa = normalize_u(eigenvectors_a[:, 1], r)
87     new_u1sb = normalize_u(eigenvectors_b[:, 0], r)
88
89     e1sa, e2sa = eigenvalues_a[0], eigenvalues_a[1]
90     e1sb = eigenvalues_b[0]
91
92     delta_e = abs(e1sa - E_1sa_history[-1]) if E_1sa_history else 999
93     # Convergence check on the energy
94     E_1sa_history.append(e1sa)
95     E_1sb_history.append(e1sb)

```



```

95     E_2sa_history.append(e2sa)
96
97     if delta_e < TOLERANCE:
98         # Total Energy Calculation (0.5 * (E_core + E_orbital))
99         H_core = T + V_nuc
100         E_core = (np.dot(u1sa, H_core.dot(u1sa)) +
101                  np.dot(u1sb, H_core.dot(u1sb)) +
102                  np.dot(u2sa, H_core.dot(u2sa))) * dr
103         E_total = 0.5 * (E_core + e1sa + e1sb + e2sa)
104         break
105         # Calculation for the energy if convergence is reached
106     u1sa, u1sb, u2sa = new_u1sa, new_u1sb, new_u2sa
107     # Sets the new iteration up

```